

HAT-P-19b

R-1069

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_241125 · created 2026-07-30T13:35:50+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460639.684 +/- 0.0052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.181 +/- 0.052
Transit depth (Rp/Rs)^2	3.29 +/- 1.88 [%]
Orbital Inclination (inc)	89.89 +/- 1.59
Airmass coefficient 1 (a1)	1.034 +/- 0.068
Airmass coefficient 2 (a2)	-0.03 +/- 0.1
Scatter in the residuals of the lightcurve fit is	6.64 %
Best Comparison Star	#2 - [462, 301]
Optimal Aperture	3.27
Optimal Annulus	12.53
Transit Duration (day)	0.0922 +/- 0.0093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.181 ± 0.052 vs 0.1418 (deviation 0.75σ)
Duration fit vs published	0.0922 d vs 0.1196 d (deviation 2.94σ)
Mid-time O–C	-10.3 min
Depth SNR	3.5
Residual scatter	6.64 %

Light curve

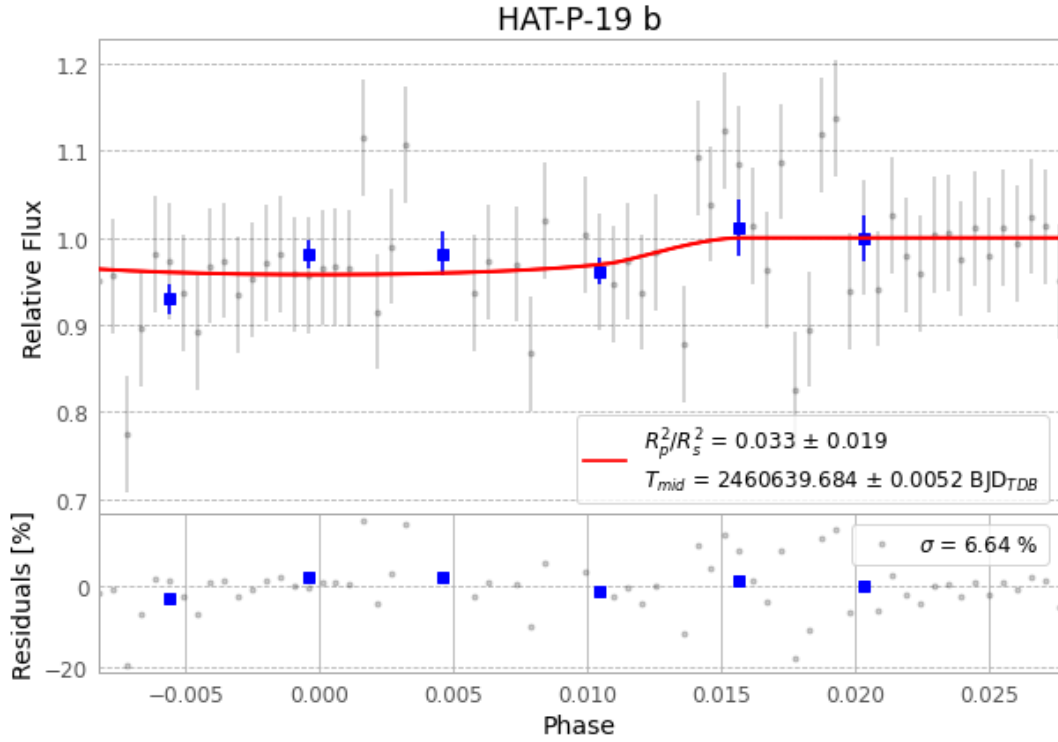


Figure 1 — FinalLightCurve_HAT-P-19 b_2024-11-24.png (SHA-256 1dbb12720d38c03f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [385, 257], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1f739c6e451e9a9c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

70 FITS frames (46 MB), HATP-19241125033015.FITS → HATP-19241125065716.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-241125.log (3bcf655883bb...), FinalLightCurve_HAT-P-19 b_2024-11-24.png (1dbb12720d38...), FinalLightCurve_HAT-P-19 b_2024-11-24.csv (65181783b50b...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 13:35Z.